Assignment 2: Git and Stream Temperature - Data Download

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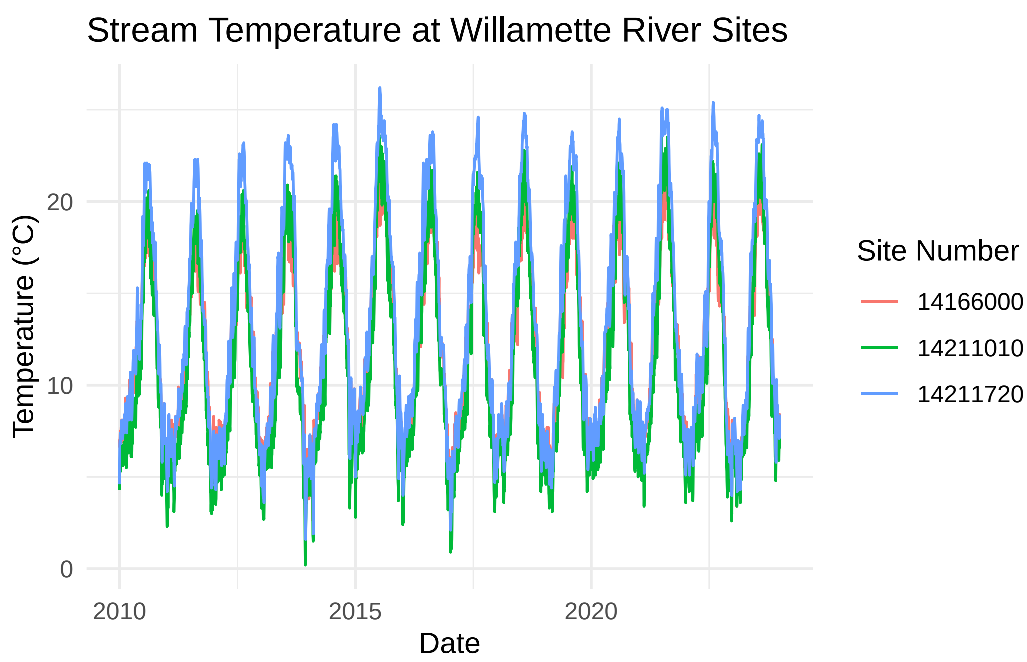
```r
library(dataRetrieval)

# Note the "USGS-" prefix added to each ID
siteNumbers <- c("USGS-14211720", "USGS-14211010", "USGS-14166000")

temp_data <- read_waterdata_daily(
  monitoring_location_id = siteNumbers,
  parameter_code = "00010",           # Water temperature
  statistic_id = "00003",             # Mean daily value
  time = c("2010-01-01", "2023-12-31"),
  skipGeometry = TRUE                 # Standard practice unless you need map coordinates
)
```

```r
library(lubridate)
library(tidyverse)

temp_data <- temp_data %>%
  select(
    agency_cd = monitoring_location_id,
    site_no = monitoring_location_id,
    Date = time,
    temp_C = value
  )

# Clean site names
temp_data$site_no <- gsub("USGS-", "", temp_data$site_no)

# Ensure Date is Date format
temp_data$Date <- as.Date(temp_data$Date)
```

```r
# Extract date info
temp_data$month <- month(temp_data$Date, label = TRUE)
temp_data$year <- year(temp_data$Date)

# Plot
ggplot(temp_data, aes(x = Date, y = temp_C, color = site_no)) +
  geom_line() +
  labs(title = "Stream Temperature at Willamette River Sites",
       x = "Date", y = "Temperature (°C)", color = "Site Number") +
  theme_minimal()
```




```r
download_data <- TRUE

if(download_data){
  temp_data <- read_waterdata_daily(
    monitoring_location_id = siteNumbers,
    parameter_code = "00010",
    statistic_id = "00003",
    time = c("2010-01-01", "2023-12-31"),
    skipGeometry = TRUE
  )
```

```r
  # cleaning steps here...

  write.csv(temp_data, "data/temp_data.csv", row.names = FALSE)

} else {
  temp_data <- read.csv("data/temp_data.csv")
}
```